

Probabilistic Reachability Bounds for Multi-Phase Entry under Process and Measurement Noise

Ethan Knox

Independent Researcher, Champaign, IL, USA

2026

Abstract

Keywords: stochastic reachability; Fokker-Planck equation; occupation measures; supermartingales; stochastic barrier certificates; concentration of measure; hybrid systems; atmospheric entry

MSC 2020:

Contents

1	Introduction	4
1.1	The price of hard bounds	4
1.2	Process noise and measurement noise	4
1.3	Thesis statement	4
1.4	Outline of contributions	4
2	The Stochastic Multi-Phase Model	5
2.1	Ito dynamics per mode	5
2.2	Where the diffusion comes from	5
2.3	Guards under noise: hitting times and boundary behavior	5
2.4	The observation model with unbounded noise	5
3	From Liouville to Fokker-Planck	6
3.1	The stochastic occupation measure	6
3.2	The generator and the weak forward equation	6
3.3	Per-mode measures and stochastic linkage	6
4	Concentration Against Diffusion	7
4.1	Two competing limits	7
4.2	Diffusive spreading of the boundary layers	7
4.3	The critical noise scale	7
4.4	Where the real atmosphere sits	7
4.5	Consequence for the certificate	7
5	Probabilistic Certificates	8
5.1	Why containment is unavailable	8
5.2	Supermartingales and the exit probability	8
5.3	Deriving the certificate	8
5.4	Numerical cross-check by sum-of-squares	8
5.5	Trading probability against volume	8
5.6	Set erosion and probabilistic tubes	8
6	Sequential Conditioning under Unbounded Noise	9
6.1	From restriction to reweighting	9
6.2	The conditioned stochastic occupation measure	9
6.3	Contraction in probability	9
6.4	Process noise against measurement information	9
7	Verification and Results	10
7.1	Recovering the deterministic limit	10
7.2	Where the real atmosphere sits	10
7.3	Exit probability against Monte Carlo	10
7.4	Probability against volume	10
7.5	Contraction under noisy returns	10
7.6	The three-way balance	10
7.7	Reproducibility environment	10

8	Discussion	11
8.1	Limitations	11
8.2	What the three papers establish together	11
8.3	Extensions	11
A	Ito Calculus Across Guards	13
A.1	The generator of the hybrid diffusion	13
A.2	Hitting times and the boundary	13
A.3	The deterministic limit	13
B	Diffusion Models from Atmospheric Data	13
B.1	Density variability	13
B.2	Wind variability	13
B.3	Polynomial form of the diffusion coefficient	13
C	Supermartingale Inequalities	13
C.1	The maximal inequality	13
C.2	Finite-horizon and stopped versions	13
C.3	Composition over epochs	13
D	Proof of the Concentration-Diffusion Threshold	13
D.1	Widths of the two effects	13
D.2	The threshold	13
D.3	The inflated residual	13

1 Introduction

- 1.1 The price of hard bounds**
- 1.2 Process noise and measurement noise**
- 1.3 Thesis statement**
- 1.4 Outline of contributions**

2 The Stochastic Multi-Phase Model

2.1 Ito dynamics per mode

2.2 Where the diffusion comes from

2.3 Guards under noise: hitting times and boundary behavior

2.4 The observation model with unbounded noise

3 From Liouville to Fokker-Planck

3.1 The stochastic occupation measure

3.2 The generator and the weak forward equation

3.3 Per-mode measures and stochastic linkage

4 Concentration Against Diffusion

- 4.1 Two competing limits
- 4.2 Diffusive spreading of the boundary layers
- 4.3 The critical noise scale
- 4.4 Where the real atmosphere sits
- 4.5 Consequence for the certificate

5 Probabilistic Certificates

- 5.1 Why containment is unavailable
- 5.2 Supermartingales and the exit probability
- 5.3 Deriving the certificate
- 5.4 Numerical cross-check by sum-of-squares
- 5.5 Trading probability against volume
- 5.6 Set erosion and probabilistic tubes

6 Sequential Conditioning under Unbounded Noise

6.1 From restriction to reweighting

6.2 The conditioned stochastic occupation measure

6.3 Contraction in probability

6.4 Process noise against measurement information

7 Verification and Results

- 7.1 Recovering the deterministic limit
- 7.2 Where the real atmosphere sits
- 7.3 Exit probability against Monte Carlo
- 7.4 Probability against volume
- 7.5 Contraction under noisy returns
- 7.6 The three-way balance
- 7.7 Reproducibility environment

8 Discussion

8.1 Limitations

8.2 What the three papers establish together

8.3 Extensions

References

A Ito Calculus Across Guards

A.1 The generator of the hybrid diffusion

A.2 Hitting times and the boundary

A.3 The deterministic limit

B Diffusion Models from Atmospheric Data

B.1 Density variability

B.2 Wind variability

B.3 Polynomial form of the diffusion coefficient

C Supermartingale Inequalities

C.1 The maximal inequality

C.2 Finite-horizon and stopped versions

C.3 Composition over epochs

D Proof of the Concentration-Diffusion Threshold

D.1 Widths of the two effects

D.2 The threshold

D.3 The inflated residual